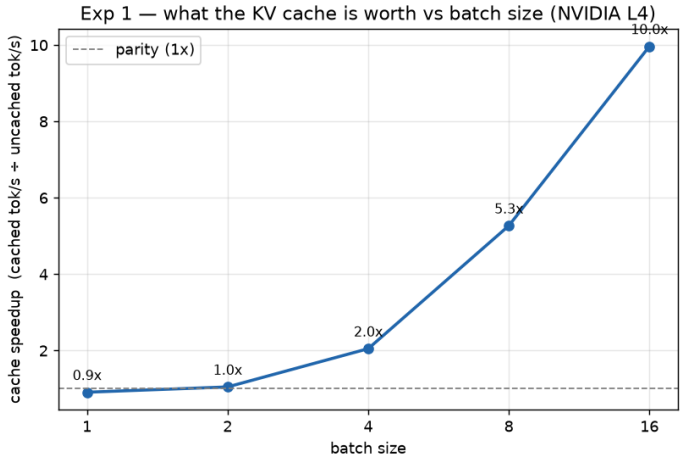
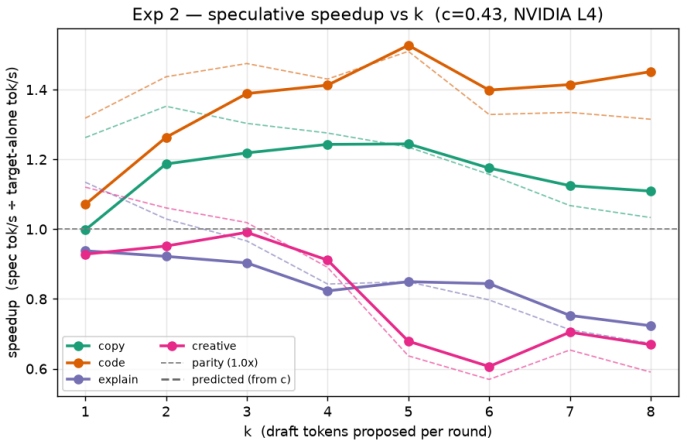
What Each Inference Trick Is Worth

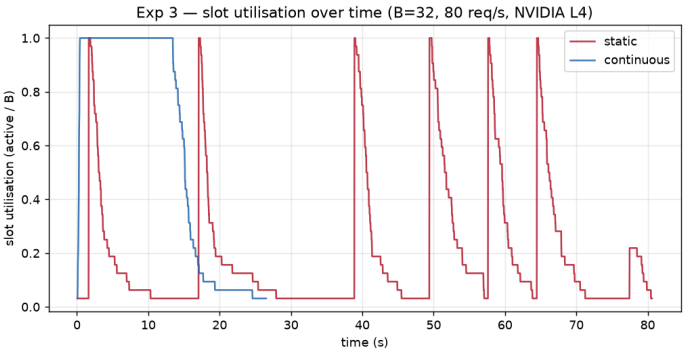
One pipeline, four controlled experiments — what each serving optimization is really worth on an NVIDIA L4.

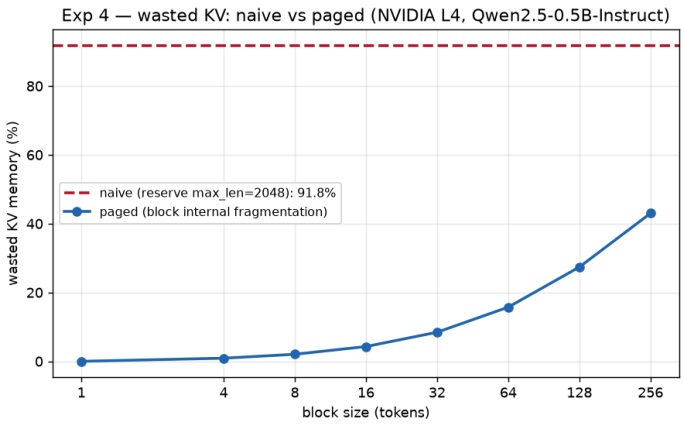
All numbers are on NVIDIA L4 (bf16), synchronised, warmed up, and repeated with their spread. Trust: a validation gauntlet (8/8 pass) shows a single un-synchronised matmul under-reports its time by 334x, so every timer here calls cuda.synchronize(); the CUDA-event and perf_counter clocks agree to 0.7%, and measured FP32 throughput stays under the card's roofline. Both sides of every comparison were checked to emit the same tokens before any speedup was claimed.

Exp	Toggle	Headline measurement
1 KV cache	recompute vs cache	batch 1: 0.91x (60x more math, same time); batch 16: 10.0x
2 Spec decode	target alone vs draft+verify	c=0.43 (draft launch-bound); best speedup 1.53x; exact modulo bf16 ties
3 Cont. batching	static vs continuous	continuous 7.5 vs static 2.5 req/s; p95 14s vs 68s
4 PagedAttention	reserve-max vs paged blocks	naive waste 92%; paging fits ~12x more seqs









Exp 1 *Expected the cache to be a big win everywhere.*
At batch 1 it is a wash (0.91x) — the uncached loop does 60x more arithmetic but the GPU was memory-bound and idle, so the extra math is free; the cache only wins as the batch fills the ALUs.

Exp 2 *Expected a 15x-smaller draft to give a big speedup.*
Naive it was a slowdown: the 0.5B draft is kernel-launch-bound (step time tracks layers, not params), so c=0.43 and no acceptance rate can win. The diagnosis, not the mechanism, is the work.

Exp 3 *Expected static batching to be slow.*
It is not slow, it is idle: the batch drains to almost empty while one long request finishes, and continuous batching holds the slots full for ~3x the throughput.

Exp 4 *Expected paging to save some memory.*
Naive reserving max_len wastes 92% of KV against a real length mix; paging fits ~12x more sequences, and that concurrency is where memory becomes speed.

Setup, diagnosis & cost

Exp 1: gpt2 (125M), a 512-token prompt, 64 generated tokens, batches 1-16, greedy. Exp 2: Qwen2.5-7B-Instruct (target, 28 layers, 60 ms/step) + Qwen2.5-0.5B-Instruct (draft, 24 layers, 25 ms/step), 128 tokens, greedy -> measured c = 0.43. I tried to bring c down by CUDA-graphing the draft (torch.compile reduce-overhead, then manual capture), but both crashed with device-side asserts on this torch 2.13 / transformers 5.15 stack, so I report the honest c = 0.43 rather than a number I could not reproduce. Exp 3 & 4: Qwen2.5-0.5B-Instruct (grouped-query attention, 2 kv-heads -> 12 KB KV per token). Exp 3 is a discrete-event simulation over the real measured decode-step times (Poisson arrivals, heavy-tailed output lengths); Exp 4's paged concurrency is confirmed on the real vLLM engine (15,220 sequences vs naive's 801). All bf16 on one NVIDIA L4 via Modal; total spend ~$2.6.

The one measurement that changed how I think about serving:

At batch 1 the uncached loop computes 60x more token positions than the cached loop and finishes in the same wall-clock time. Serving a single stream is memory-bound: the arithmetic units sit idle waiting on weights, so 'wasted' compute is free — until you add enough concurrent sequences that arithmetic becomes the wall. Every one of these four techniques is really about the same thing: keeping the expensive, bandwidth-limited GPU busy with useful work instead of waiting or reserving.